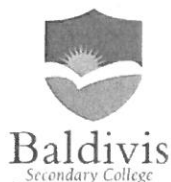


Name: _____

Date: _____

Class: _____

**Year 8 Mathematics****Mini-Test 2, 2019****Topic – Perimeter, Area and Volume****41**

Total Time: 40 min
Weighting: 5 %
Equipment: 1 page of notes and a calculator

-1 for no units.
 overall.

Question 1**3 marks**

Convert the following units:

a. $27\text{mm} = \underline{2.7} \text{ cm} \div 10$

b. $0.00084\text{m}^2 = \underline{8.4} \text{ cm}^2 \times 100^2$

c. $7.26\text{m}^3 = \underline{7,260,000} \text{ cm}^3 \times 100^3$

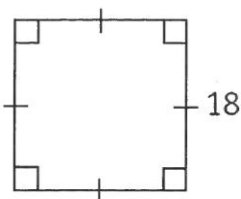
Question 2**10 marks**

a. What is the definition of perimeter: the distance along the boundary of the shape ✓

b. Find the perimeter of the following shapes. All units are in centimetres:

a.

Working Out:

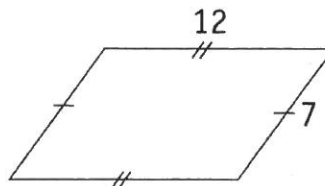


$$18 \times 4 \\ = 72 \quad \checkmark$$

Answer: 72cm ✓

b.

Working Out:

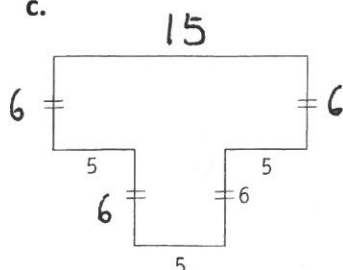


$$12 + 12 + 7 + 7 \\ = 38 \quad \checkmark$$

Answer: 38cm ✓

c.

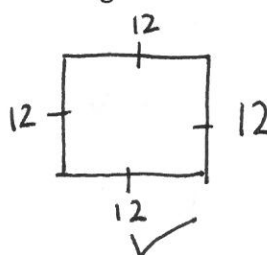
Working Out:



$$15 + 15 + 12 + 12 \\ = 54 \quad \checkmark$$

Answer: 54cm ✓

d. If the perimeter of a square is 48 centimetres, find the length of each side.



$$p = 48 \\ 48 \div 4 \quad \checkmark \\ = 12\text{cm}$$

Answer: 12cm ✓

Question 3

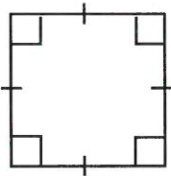
11 marks

a. What is the definition of area: the inside space in a 2D shape ✓

b. Find the area of the following shapes.

a.

Working Out:



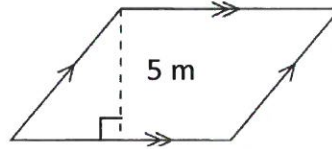
6.1 cm

$$6.1 \times 6.1 = 37.21 \quad \checkmark$$

Answer: 37.21 cm² ✓

b.

Working Out:



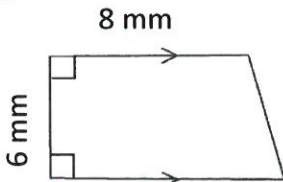
14 m

$$14 \times 5 = 70 \quad \checkmark$$

Answer: 70 m² ✓

c.

Working Out:



1.2 cm

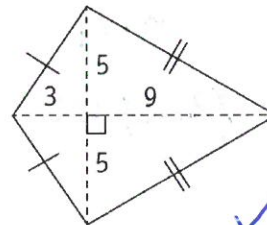
✓ 12 mm

$$0.5 \times (12 + 8) \times 6 = 60 \text{ mm}^2 \quad \checkmark$$

Answer: 60 mm² ✓

d. (Units is cm)

Working Out:



$$x = 5 + 5 = 10$$

$$y = 3 + 9 = 12$$

$$0.5 \times 10 \times 12 = 60 \quad \checkmark$$

Answer: 60 cm² ✓

Question 4

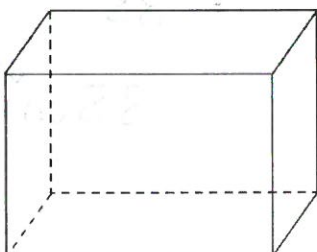
5 marks

a. What is the definition of volume: The space inside of a solid 3D shape ✓

b. Find the volume of the following shapes.

a.

Working Out:



11 cm

9.5 cm

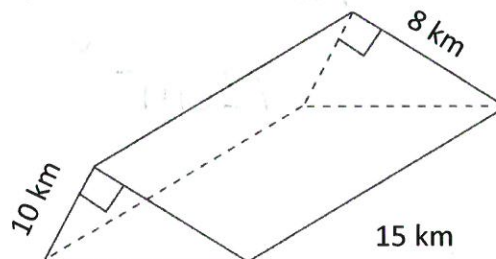
15 cm

$$15 \times 9.5 \times 11 \quad \checkmark$$

Answer: 1567.5 cm³ ✓

b.

Working Out:

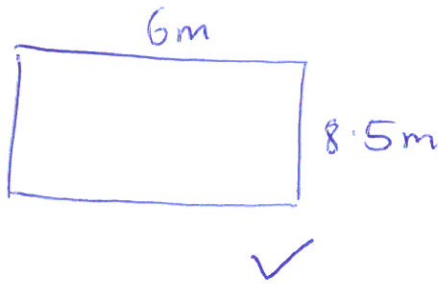


$$0.5 \times B \times H \times L \quad \checkmark$$

$$0.5 \times 10 \times 8 \times 15$$

Answer: 600 km³ ✓

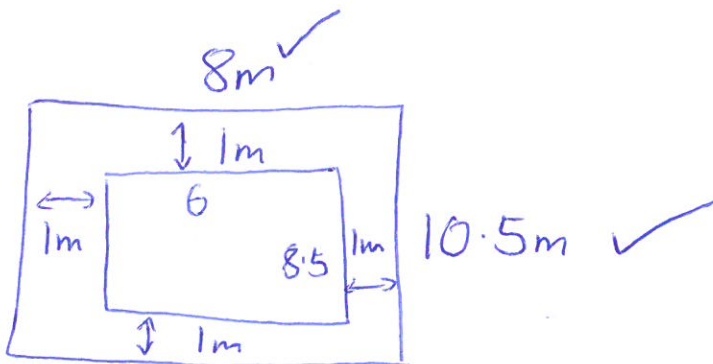
- a. A rectangular swimming pool is 6 m $\times$ 8.5 m.
i. What is the area of the swimming pool's surface?



$$6 \times 8.5 \text{ m} \checkmark$$

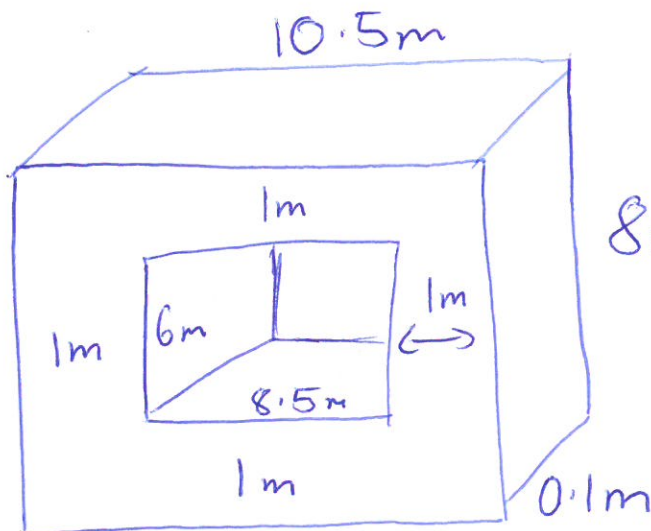
$$= 51 \text{ m}^2 \checkmark$$

- b. The swimming pool is surrounded by a concrete path that is 1 m wide and 10 cm deep
i. What are the dimensions of the swimming pool including the path?



$\checkmark$ one for overall diagram.

- ii. What volume of concrete will be needed to make the path?



$\checkmark$ diagram

$$10.5 \times 8 = 84 \text{ m}^2 \checkmark$$

$$6 \times 8.5 = 51 \text{ m}^2 \checkmark$$

$$\text{Area} = 84 - 51 = 33 \text{ m}^2 \checkmark$$

$$\text{Volume} = 33 \times 0.1 \checkmark$$

$$= 3.3 \text{ m}^3 \checkmark$$

End of Test

